AIDAN ACQUAH

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PROFILE

Highly skilled Software Engineer and Data Scientist with a deep specialisation in Machine Learning and Health Data Science. I am currently a Research Fellow at University College London, holding a DPhil in Engineering Science (Health Data Science) from University of Oxford, and a First-Class Master's in Biomedical Engineering (MEng) from Imperial College London. My research has focused on the use of wearable sensors and AI for the prediction and prevention of disease.

Prior experience includes working as an Analyst Programmer at Tessella Ltd, where I developed production-grade software for major clients like BP, Rolls-Royce, and Unilever. I have also made significant contributions to open-source health data projects, including widely used Python packages for data analysis.

EDUCATION

University College London **Research Fellow | May 2026 – Present**

- Development of behaviour change interventions using smart devices
- Lead wearable data collection projects, supervising Master's and Doctoral students

University of Oxford **Postdoctoral Researcher | April 2025 – April 2026**

- Produced publications, posters and conference presentations in wearable device research
- Worked on a variety of collaboration projects for improving machine learning algorithms

University of Oxford **DPhil in Engineering Science (Health Data Science) | 2020 – 2025**

- Researched predictive analytics using wearable sensor data, with applications in Parkinson's disease detection
- Handled processing of large datasets using Slurm.
- Applied deep learning (ResNet-18, self-supervised models) and statistical analysis (Cox regression) in Python and R.
- Developed algorithms for activity recognition and gait analysis using wrist-worn accelerometers

Imperial College London **MEng in Biomedical Engineering (First Class) | 2013 – 2017**

- Attended classes in digital signal processing, programming and machine learning

EXPERIENCE

GSK plc **Researcher (Machine Learning) | March 2023 – June 2023**

- Developed machine learning models for digital biomarker discovery in a global research setting.
- Conducted analyses on health datasets to uncover predictive insights into disease progression.
- Collaborated with international teams to develop AI-driven healthcare solutions.

Tessella Ltd **Analyst Programmer | September 2017 – August 2020**

- Designed and built data-driven software solutions for clients in energy, pharmaceuticals, and manufacturing.
- Developed production-grade applications using .NET, Angular, React, and Python.
- Delivered high-performance machine learning models and data pipelines for mission-critical systems.
- Worked in agile teams, collaborating directly with stakeholders to define project goals and technical solutions.

SELECTED PUBLICATIONS

- Acquah et al. (2025). Daily steps are a predictor of, but perhaps not a risk factor for Parkinson's Disease: findings from the UK Biobank. [npj Parkinson's Disease]
- Acquah et al. (2026). ActiNet: An Open-Source Tool for Activity Intensity Classification of Wrist-Worn Accelerometry Using Self-Supervised Deep Learning. [arXiv preprint]
- Acquah et al. (2026). Evaluating Open-Source Wrist-Worn Accelerometer Models for Sedentary Time Detection Against Thigh-Worn Accelerometer Data. [medrXiv preprint]
- Yuan et al. (2024). Self-supervised learning for human activity recognition using 700,000 person-days of wearable data. [npj Digital Medicine]
- Chan et al. (2024). CAPTURE-24: A large dataset of wrist-worn activity tracker data collected in the wild for human activity recognition. [Scientific Data]
- Small et al. (2024). Self-supervised machine learning to characterize step counts from wrist-worn accelerometers in the UK Biobank. [Medicine and Science in Sports and Exercise]

CONFERENCE PRESENTATIONS

- Acquah & Broomberg et al. (2026). Evaluating Open Source Wrist Worn Accelerometer Models for Sedentary Time Classification. Oral presentation, ICAMPAM 2026.
- Acquah et al. (2025). Walking recognition in Parkinson's disease populations using wrist-worn accelerometers. Oral presentation, ICAMPAM 2024. Abstract in *Journal for the Measurement of Physical Behaviour*.
- Acquah et al. (2025). Characterising of Parkinson's disease-like walking using wrist-worn accelerometers. Poster presentation, UbiComp / ISWC 2025. Extended abstract in *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing*.

AWARDS

- Early Career Researcher Oral Award. ICAMPAM 2026.

OTHER

- Taught graduate-level time-series machine learning and programming (C++, R, Python) at the University of Oxford (2022–2025), also developing course materials for international researchers.
- Languages: English (Fluent), French (B2), Twi (Conversational)
- Hiking, including summiting Mt. Kilimanjaro in 2024.
- Board games, including Dungeons & Dragons.
- Volunteering, at a food bank